Arrival-Process Microphysics and Instrument Validity in a Mission-Level Simulator

Note

This is the companion to SANDTABLE's main report. The main paper's claim is that span-of-control collapse in a supervisory console is congestion, that closed forms locate it, and that software moves it. Establishing that claim required a second body of work, on how an arrival process at a console is measured and on what a burstiness statistic can and cannot see, which is longer than the claim it supports and is about a different subject. It is here rather than there. Every number below is generated by the same scripts, from the same battery logs, under the same registration discipline, and the registered-claim ledger in the main paper covers both.

1 The retry cohort, and what a mean could not tell us about it

The main paper keeps one sentence from this material: a fleet retrying on a jittered schedule presents the console with an arrival pattern statistically indistinguishable from a fleet that never retries at all, while the same fleet on a fixed schedule presents one more than twice as crowded at the largest span. What follows is how that sentence was arrived at, including the two readings of it that were published and withdrawn.

That number is smaller than the one this paragraph carried in an earlier draft, and the difference is worth a sentence because it is the same measurement problem as Section **??**'s. The claim was stated on operator goodput, and a jittered hold changes when a rebuffed vehicle comes back, which changes how often it is refused again, which moves the denominator. Read on the two quantities that have no denominator, the recommendation is narrower and clearer. Of the 6 mean-matched cells, 2 gate on mission success, both at the span where the queue sits near its threshold. On the absolute service rate the effect is unambiguous and small: jitter raises what the console delivers by +7% to +27% across spans and channels, at $|t|$ up to 21.7 with almost every block agreeing, and the largest of those gains anywhere is +0.0065 decisions per step, which is 4.3% of what one console can deliver and does not clear a materiality floor set at six percent of that capacity. So the honest statement of the recommendation is that a randomized hold is worth a few points of mission outcome where the queue is near its threshold, at a cost of one random draw on the asset side, and that it is not measurable as console throughput. The two readings are reported together because a reader deciding whether to implement it needs the regime as well as the number. Across every cell that paragraph reads, the console is busy 0.42 to 0.91 of the time and delivers 4 to 37 percent of its capacity as service anyone is still waiting for. That gap is not slack, and it is the whole subject of Section **??**: the console is working, and most of what it produces arrives after its requester has given up. A relative gain of a seventh, measured against a base that small, is a smaller failure rather than an operational gain, which is why the floor is set against what a console can deliver rather than against what it happens to be delivering.

We had expected that to be a fact about the link. Vehicles knocked offline by a common fade come back when it lifts, so a fixed hold should march them into the console together at the recovery edge, and a bursty channel should therefore be where jitter pays. The prediction was that jitter would lower the burstiness of the delivered request stream, measured as its Fano factor. We first measured it in windows the size of the channel's own dwell, found no pair with a gated reduction, the largest being -0.0911 at $t = -1.2$, and reported that as a refutation. It is not one. The window had been chosen by an argument about where the channel's structure lives rather than by measuring where the statistic works, and those give opposite answers. Benched against an injected cohort of known size, the estimator recovers 1.307 of it at the dwell window at t near 4, against 0.997 at t above 376 at a window one service time wide. A null read through a fifth of the available power is not evidence of absence. We therefore re-ran the same arms with the window taken from the bench's own power curve rather than from the channel, and registered the prediction again before looking at any number the re-run produced. It is confirmed in 6 of 6 mean-matched pairs on the bursty link, by -0.32 to -1.53 against fixed-hold Fano factors of up to 2.99, which is -20% to -53% of the burstiness the fixed hold produces, at $|t|$ from 27 to 89 with 32 of 32 blocks agreeing in every pair. The multiplicity statistic, which unlike a variance can separate many small clusters from few large ones, agrees in 6 of 6. The published null was the window and nothing else. The control we had included to be discarded is the one that carries the result, though not in the way we first read it. On a memoryless link there is no dwell to synchronize to, so jitter should have bought nothing. It buys more there than on the bursty link, not less: at six assets the delivered service rate rises +12% with no dwell present against +8% with one ($t = 13.8$ and 5.3, 32 and 26 of 32 blocks). And the Fano reduction appears there too, in 3 of 4 pairs, smaller in absolute and relative terms than on the bursty link (-0.23 to -0.35, -18% to -24%). What the control establishes is therefore not that the channel is uninvolved. It is that a retry cohort forms, and costs the console real throughput, in a link with no structure at all for it to form around. A common outage puts several vehicles into the queue together; a deterministic hold is what keeps them together on the next attempt, and the one after, and that second part needs no channel. The bursty link has more clustering to dissolve because its gate adds its own on top of the retry's, which is why the same intervention removes more of it there. Four statistics now say the same thing from different directions: the raw collision multiplicity, the multiplicity net of the arrival rate that would produce it under Poisson, the Fano factor at the mechanism's own timescale, and the residual that survives jitter at matched delivered rate. All four make the bursty link the more clustered one and jitter the larger correction on it.

The measurement that settles what a randomized hold removes is a control we did not run until late, and running it made most of what preceded it unnecessary. Every earlier reading compared a jittered arm against a calculated floor: the crowding a perfectly scattered stream would show, or a two-state source's, or a gated source's. Three references gave three answers and we argued about which was right. The reference a fleet that retries should be compared against is a fleet that does not, run under everything else identical, and it cancels every nuisance parameter the calculated floors disagreed about.

Sweeping the fade length separates what each part contributes, and it took re-running the sweep with each arm's own delivered rate logged before any of it could be read, because three of the four terms are levels and a level needs a computed reference. Over 10 fade lengths spanning a 240-fold range, the crowding decomposes into four terms and only one of them belongs to the channel. What a gated link contributes by itself, computed in closed form from its own dwell and the measured delivery rate, runs 0.964 to 1.031. A fleet whose rebuffed vehicles do not come back sits 0.056 above that, and sits there at every fade length: across the whole 240-fold range that term varies by 0.0115, which is to say it does not vary. A fixed hold adds a retry cohort on top, and *that* is the term the fade governs, growing from 0.143 to 0.325 and rising at 9 of 9 steps

with $|t|$ never below 30. A randomized hold of the same mean adds nothing at all: its excess over the channel's own contribution is the no-retry fleet's, to within the spread just quoted. That last term is now measured against the control directly rather than inferred: the jittered fleet against a fleet that never retries, at each of the 10 fade lengths on its own blocks, gates at 0 of them and never differs by more than 1.1%. A mean cannot say whether a longer fade makes the console collide more often or makes each collision bigger, and those are different design problems. The share of service windows that saw two or more distinct vehicles answers the first, the share that saw three or more the second, and against the no-retry control they answer differently. A fixed hold raises the two-or-more share by +0.0364 at the shortest fade and +0.0499 at the longest, a growth factor of 1.4 across a 240-fold range, which is to say the fade barely changes how often the console is collided with. Over the same range it raises the three-or-more share from +0.0658 to +0.1517, a factor of 2.3, and the mean excess grows by 2.5, which is the same factor. All three gate at all 10 fade lengths with $|t|$ never below 12. A longer fade does not produce more collisions at the console; it produces deeper ones, and the depth is what a mean was reporting all along. Deeper collisions are the worse failure for a console, because the vehicles at the back of a larger cohort are the ones whose patience expires while they wait. We report the decomposition in that form because an earlier draft reported a single residual, and the disagreements it caused were all disagreements about which of these four things the word named. The one that grows with the fade is the retry cohort and it is removed entirely by randomizing the hold. The one that survives randomization is not a retry phenomenon, does not respond to the channel, and is what a fleet that never retries already has. Separating the two components costs no new computation, because the two links are sections of one battery run and block i carries the same seed in each, so the cross-channel difference in differences is itself paired. Measured where no dwell exists to confound it, self-collision is worth +0.0400 of operator goodput at five assets and +0.0490 at six ($t = 31.0$ and 30.6, 32 of 32 blocks in both). What the dwell contributes on top is +0.0112 at five and -0.0248 at six: neither reaches the minimum practical effect, and the two disagree in sign. The whole of the benefit that survives the gate is vehicles desynchronizing from each other, and the recommendation is correspondingly indifferent to what the link is doing.

And an earlier version of this section reported a residual that survived jitter and grew with the channel's dwell. There is no such residual. What we were measuring was the difference between two calculated references, and it is withdrawn along with the dwell sweep that appeared to support it, which was read against a null of exactly one where a gated stream's true value runs from 0.957 at an outage of 1 step to 1.044 at 240 and moves as much again with the delivered rate. Both were levels compared against constants that are functions, which is the defect this paper has now made five times and the reason Section **??** treats instruments as registrable claims.